

INSTITUT UNIVERSITAIRE DES SCIENCES

IUS- FST-L3 Administration Systemes

TP N°7

Nom : EXUME

Prénom : Billy Rolph

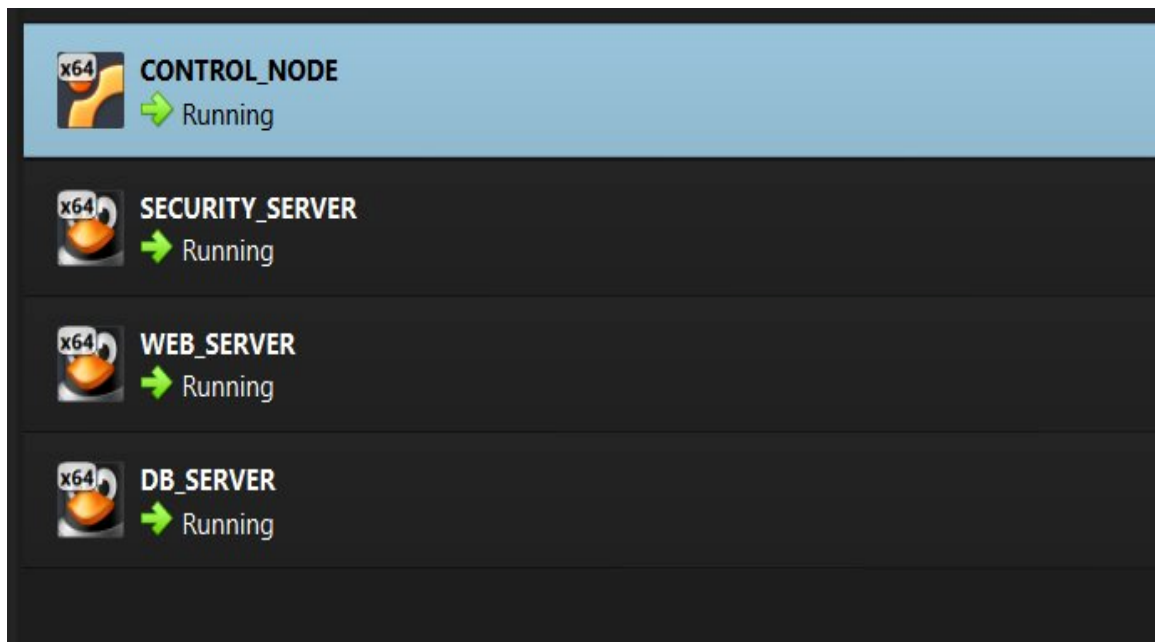
Faculté : FST-INFORMATIQUE-L3

Prof : Ismael Saint Amour

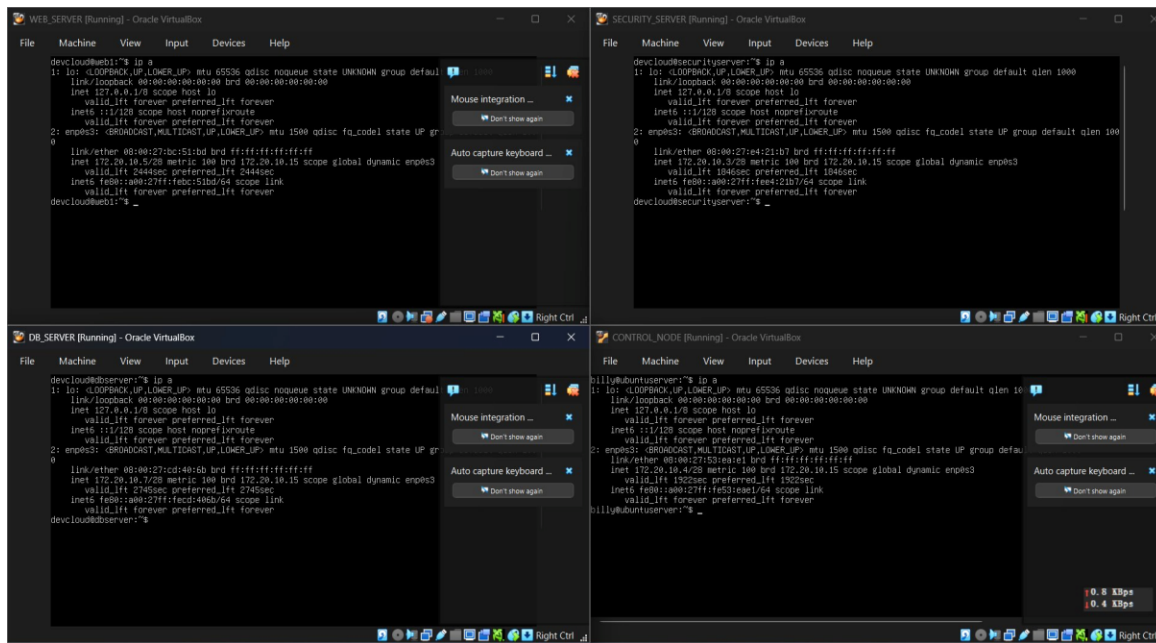
Date :19 AVRIL 2026

Introduction : Dans le cadre de ce TD, nous avons travaillé sur l'automatisation du déploiement d'une infrastructure avec Ansible pour la startup fictive DevCloud. L'objectif principal était de paramétrer automatiquement trois serveurs distincts, un serveur web, un serveur de base de données et un serveur de sécurité. Pour cela nous avons installé Ansible, créé un inventaire, structuré le projet en rôles et utilisé des templates Jinja2 pour déployer des fichiers de configuration dynamiques.

1. Mise en place des machines virtuelles

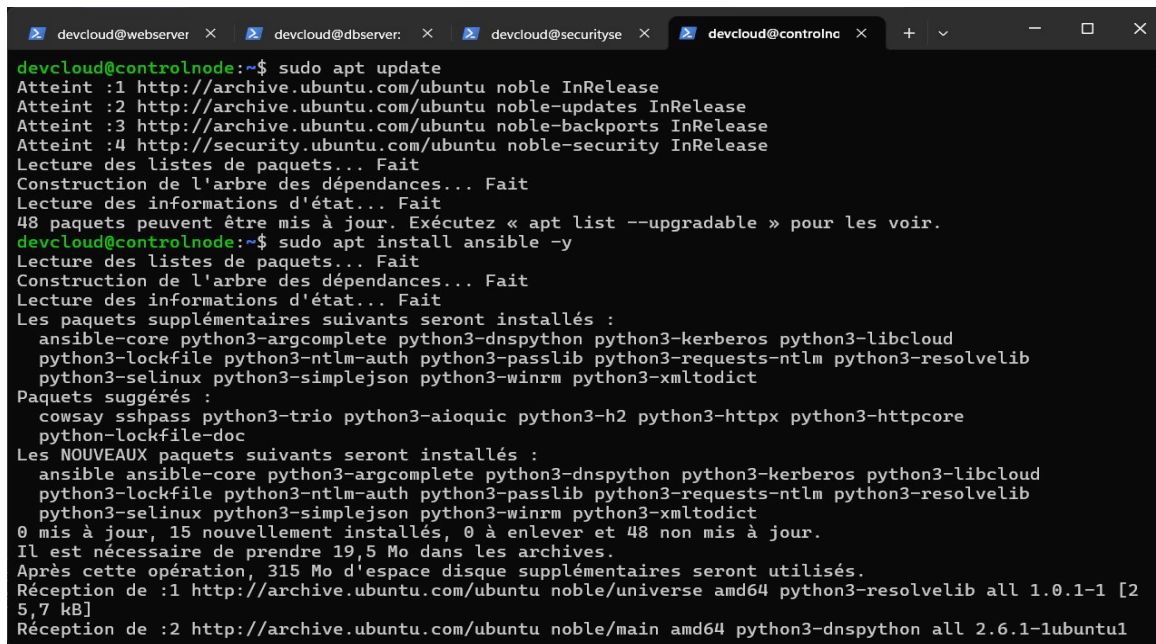


Les quatre VMs sont en cours d'exécution dans VirtualBox.



Verification des adresses IP sur chaque serveur.

2. Installation d'Ansible



Installation d'Ansible via la commande apt sur le control node.

```
devcloud@webservers x devcloud@dbserver: x devcloud@securityse x devcloud@controlnode x + - □ ×
devcloud@controlnode:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/devcloud/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/devcloud/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Mar 3 2026, 12:15:18) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
devcloud@controlnode:~$
```

Verification de la version d'Ansible installée.

3. Configuration SSH

```
devcloud@webservers x devcloud@dbserver: x devcloud@securityse x billy@controlnode: ~ x + - □ ×
billy@controlnode:~$ ssh-copy-id devcloud@172.20.10.5
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/billy/.ssh/id_rsa.pub"
The authenticity of host '172.20.10.5 (172.20.10.5)' can't be established.
ED25519 key fingerprint is SHA256:FgtTQIgKWBPSAL3/RL/ITFj/Z9kdz3gotesNQ12A/mA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
devcloud@172.20.10.5's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'devcloud@172.20.10.5'"
and check to make sure that only the key(s) you wanted were added.

billy@controlnode:~$ ssh-copy-id devcloud@172.20.10.7
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/billy/.ssh/id_rsa.pub"
The authenticity of host '172.20.10.7 (172.20.10.7)' can't be established.
ED25519 key fingerprint is SHA256:oj0tQgu1gINTfaMDmSFmKutAu/WIn1EA4NEmJXKEpl8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
devcloud@172.20.10.7's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'devcloud@172.20.10.7'"
and check to make sure that only the key(s) you wanted were added.

billy@controlnode:~$ ssh-copy-id devcloud@172.20.10.3
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/billy/.ssh/id_rsa.pub"
The authenticity of host '172.20.10.3 (172.20.10.3)' can't be established.
```

Copie des clés SSH vers les serveurs cibles.

```
billy@controlnode:~$ ssh devcloud@172.20.10.5
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

Connexion SSH reussie vers le serveur web.

```
billy@controlnode:~$ ssh devcloud@172.20.10.3
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

Connexion SSH reussie vers le serveur de securite.

```
billy@controlnode:~$ ssh devcloud@172.20.10.7
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

Connexion SSH reussie vers le serveur de base de donnees.

4. Creation de la structure du projet

```
devcloud@webserver x devcloud@dbserver: x devcloud@securityse x billy@controlnode: ~ x
billy@controlnode:~$ mkdir tp-ansible
billy@controlnode:~$ cd tp-ansible
-bash: cd: tp-ansible: No such file or directory
billy@controlnode:~$ cd tp-ansible
billy@controlnode:~/tp-ansible$ ansible-galaxy init roles/web
- Role roles/web was created successfully
billy@controlnode:~/tp-ansible$ ansible-galaxy init roles/db
- Role roles/db was created successfully
billy@controlnode:~/tp-ansible$ ansible-galaxy init roles/security
- Role roles/security was created successfully
billy@controlnode:~/tp-ansible$
```

Creation du dossier tp-ansible et des trois roles.


```
GNU nano 7.2 inventory.ini *
[web]
172.20.10.5

[db]
172.20.10.7

[security]
172.20.10.3

[all:vars]
ansible_user=devcloud

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_/ Go To Line M-E Redo
```

Contenu du fichier inventory.ini avec les trois groupes.

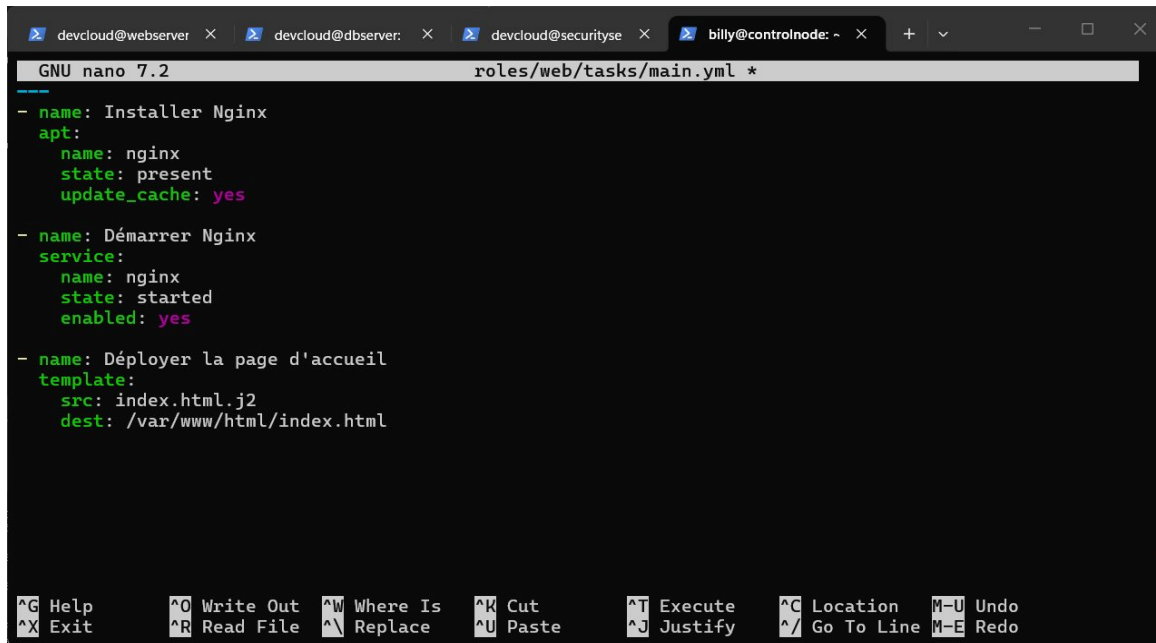
```
billy@controlnode:~/tp-ansible$ ansible all -i inventory.ini -m ping
usage: ansible [-h] [--version] [-v] [-b] [--become-method BECOME_METHOD] [--become-user BECOME_USER]
               [-K | --become-password-file BECOME_PASSWORD_FILE] [-i INVENTORY] [--list-hosts]
               [-l SUBSET] [-P POLL_INTERVAL] [-B SECONDS] [-o] [-t TREE]
               [--private-key PRIVATE_KEY_FILE] [-u REMOTE_USER] [-c CONNECTION] [-T TIMEOUT]
               [--ssh-common-args SSH_COMMON_ARGS] [--sftp-extra-args SFTP_EXTRA_ARGS]
               [--scp-extra-args SCP_EXTRA_ARGS] [--ssh-extra-args SSH_EXTRA_ARGS]
               [-k | --connection-password-file CONNECTION_PASSWORD_FILE] [-C] [-D] [-e EXTRA_VARS]
               [--vault-id VAULT_IDS] [-J | --vault-password-file VAULT_PASSWORD_FILES] [-f FORKS]
               [-M MODULE_PATH] [--playbook-dir BASEDIR] [--task-timeout TASK_TIMEOUT]
               [-a MODULE_ARGS] [-m MODULE_NAME]
               pattern
ansible: error: unrecognized arguments: -i inventory.ini

usage: ansible [-h] [--version] [-v] [-b] [--become-method BECOME_METHOD] [--become-user BECOME_USER]
               [-K | --become-password-file BECOME_PASSWORD_FILE] [-i INVENTORY] [--list-hosts]
               [-l SUBSET] [-P POLL_INTERVAL] [-B SECONDS] [-o] [-t TREE]
               [--private-key PRIVATE_KEY_FILE] [-u REMOTE_USER] [-c CONNECTION] [-T TIMEOUT]
               [--ssh-common-args SSH_COMMON_ARGS] [--sftp-extra-args SFTP_EXTRA_ARGS]
               [--scp-extra-args SCP_EXTRA_ARGS] [--ssh-extra-args SSH_EXTRA_ARGS]
               [-k | --connection-password-file CONNECTION_PASSWORD_FILE] [-C] [-D] [-e EXTRA_VARS]
               [--vault-id VAULT_IDS] [-J | --vault-password-file VAULT_PASSWORD_FILES] [-f FORKS]
               [-M MODULE_PATH] [--playbook-dir BASEDIR] [--task-timeout TASK_TIMEOUT]
               [-a MODULE_ARGS] [-m MODULE_NAME]
               pattern

Define and run a single task 'playbook' against a set of hosts
```

Test de connectivite avec le module ping d'Ansible.

5. Configuration des roles



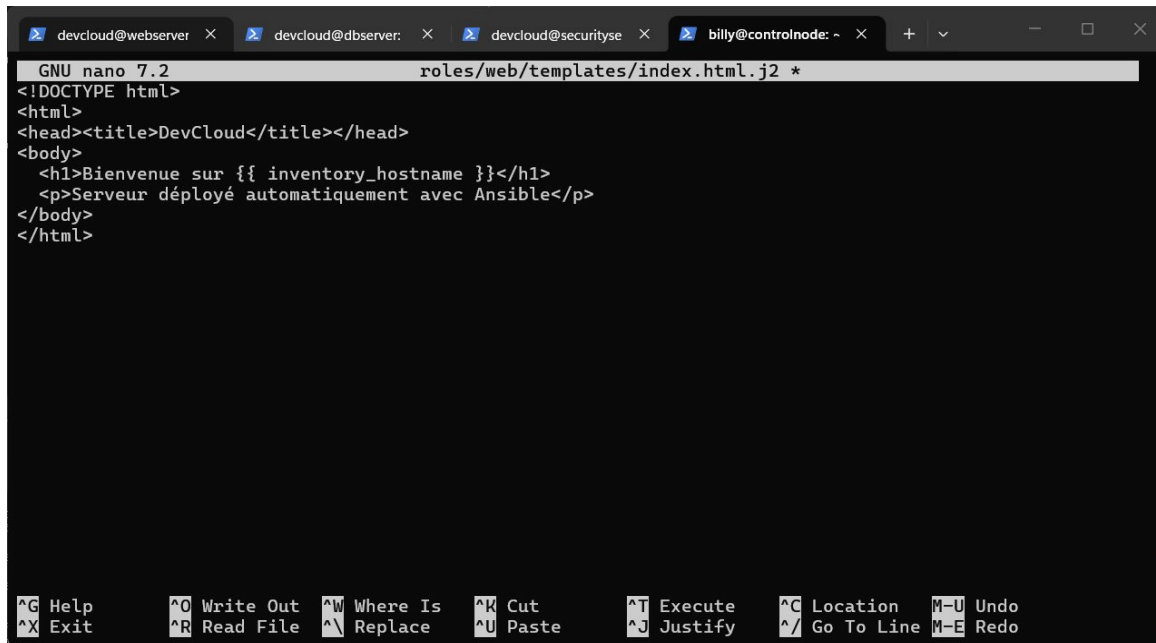
```
GNU nano 7.2 roles/web/tasks/main.yml *
- name: Installer Nginx
  apt:
    name: nginx
    state: present
    update_cache: yes

- name: Démarrer Nginx
  service:
    name: nginx
    state: started
    enabled: yes

- name: Déployer la page d'accueil
  template:
    src: index.html.j2
    dest: /var/www/html/index.html

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/_ Go To Line M-E Redo
```

Contenu du fichier main.yml du role web (Nginx).



```
GNU nano 7.2 roles/web/templates/index.html.j2 *
<!DOCTYPE html>
<html>
<head><title>DevCloud</title></head>
<body>
  <h1>Bienvenue sur {{ inventory_hostname }}</h1>
  <p>Serveur déployé automatiquement avec Ansible</p>
</body>
</html>

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/_ Go To Line M-E Redo
```

Template Jinja2 de la page d'accueil web.


```
GNU nano 7.2 roles/db/tasks/main.yml *
--
- name: Installer MySQL
  apt:
    name: mysql-server
    state: present
    update_cache: yes

- name: Démarrer MySQL
  service:
    name: mysql
    state: started
    enabled: yes

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

Contenu du fichier main.yml du role base de donnees.

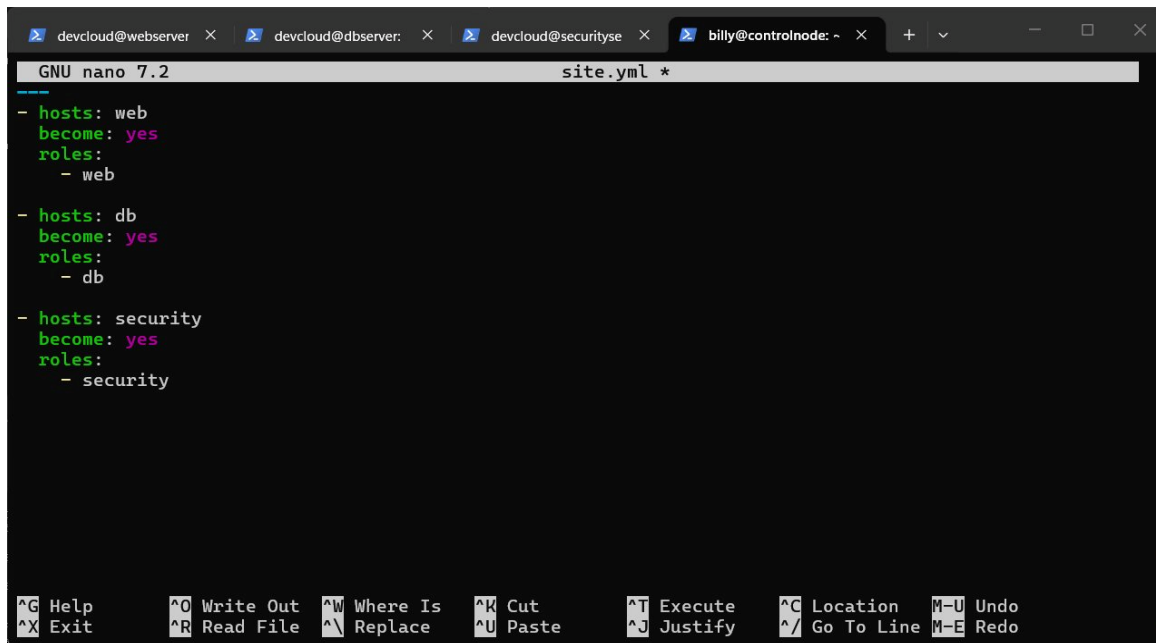
```
GNU nano 7.2 roles/security/tasks/main.yml *
--
- name: Installer UFW
  apt:
    name: ufw
    state: present

- name: Autoriser SSH
  ufw:
    rule: allow
    port: "22"

- name: Activer le firewall
  ufw:
    state: enabled
    policy: allow

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

Contenu du fichier main.yml du role securite (UFW).



```
GNU nano 7.2 site.yml *
- hosts: web
  become: yes
  roles:
    - web

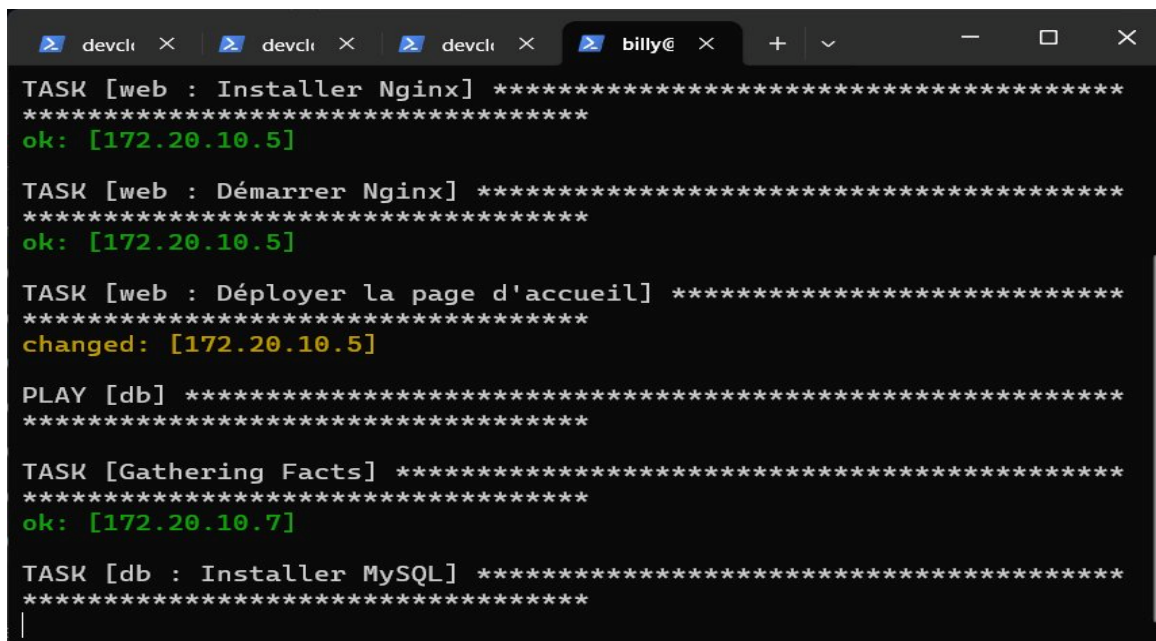
- hosts: db
  become: yes
  roles:
    - db

- hosts: security
  become: yes
  roles:
    - security

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line M-E Redo
```

Contenu du playbook principal site.yml.

6. Execution du playbook



```
devcli  x  devcli  x  devcli  x  billy@ x
TASK [web : Installer Nginx] *****
*****
ok: [172.20.10.5]

TASK [web : Démarrer Nginx] *****
*****
ok: [172.20.10.5]

TASK [web : Déployer la page d'accueil] *****
*****
changed: [172.20.10.5]

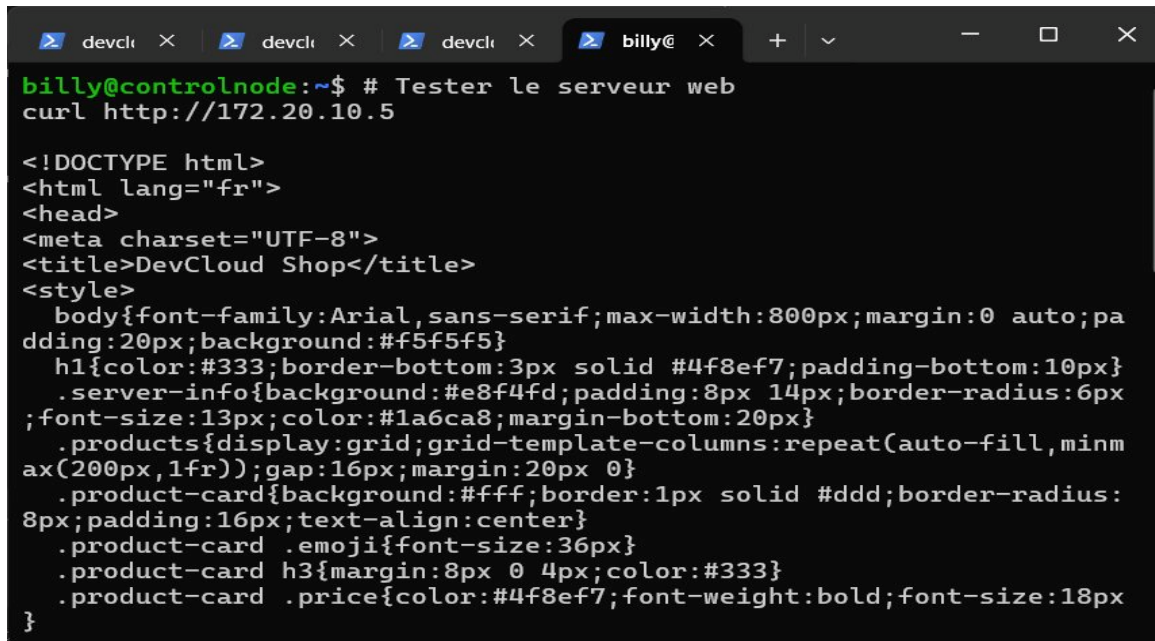
PLAY [db] *****
*****

TASK [Gathering Facts] *****
*****
ok: [172.20.10.7]

TASK [db : Installer MySQL] *****
*****
```

Execution du playbook, toutes les taches sont reussies.

7. Verification finale

A terminal window with a dark background and light-colored text. The window has four tabs at the top, the last of which is labeled 'billy@'. The terminal shows a command prompt 'billy@controlnode:~\$' followed by the command '# Tester le serveur web' and 'curl http://172.20.10.5'. The output is an HTML document. The first line is '<!DOCTYPE html>'. The second line is '<html lang="fr">'. The third line is '<head>'. The fourth line is '<meta charset="UTF-8">'. The fifth line is '<title>DevCloud Shop</title>'. The sixth line is '<style>'. The seventh line is 'body{font-family:Arial,sans-serif;max-width:800px;margin:0 auto;padding:20px;background:#f5f5f5}'. The eighth line is 'h1{color:#333;border-bottom:3px solid #4f8ef7;padding-bottom:10px}'. The ninth line is '.server-info{background:#e8f4fd;padding:8px 14px;border-radius:6px;'. The tenth line is ';font-size:13px;color:#1a6ca8;margin-bottom:20px}'. The eleventh line is '.products{display:grid;grid-template-columns:repeat(auto-fill,minmax(200px,1fr));gap:16px;margin:20px 0}'. The twelfth line is '.product-card{background:#fff;border:1px solid #ddd;border-radius:8px;padding:16px;text-align:center}'. The thirteenth line is '.product-card .emoji{font-size:36px}'. The fourteenth line is '.product-card h3{margin:8px 0 4px;color:#333}'. The fifteenth line is '.product-card .price{color:#4f8ef7;font-weight:bold;font-size:18px}'. The sixteenth line is '}'.

```
billy@controlnode:~$ # Tester le serveur web
curl http://172.20.10.5

<!DOCTYPE html>
<html lang="fr">
<head>
<meta charset="UTF-8">
<title>DevCloud Shop</title>
<style>
  body{font-family:Arial,sans-serif;max-width:800px;margin:0 auto;padding:20px;background:#f5f5f5}
  h1{color:#333;border-bottom:3px solid #4f8ef7;padding-bottom:10px}
  .server-info{background:#e8f4fd;padding:8px 14px;border-radius:6px;
;font-size:13px;color:#1a6ca8;margin-bottom:20px}
  .products{display:grid;grid-template-columns:repeat(auto-fill,minmax(200px,1fr));gap:16px;margin:20px 0}
  .product-card{background:#fff;border:1px solid #ddd;border-radius:8px;padding:16px;text-align:center}
  .product-card .emoji{font-size:36px}
  .product-card h3{margin:8px 0 4px;color:#333}
  .product-card .price{color:#4f8ef7;font-weight:bold;font-size:18px}
}
```

Resultat de la commande curl sur le serveur web.

Réponses aux questions

Pourquoi utiliser Ansible ?

Ansible est un outil d'automatisation qui permet de configurer plusieurs serveurs en même temps sans avoir à répéter les mêmes commandes manuellement sur chaque machine. Il évite les erreurs humaines, garantit que tous les serveurs ont exactement la même configuration, et utilise des fichiers YAML simples à lire et à écrire. C'est un gain de temps énorme surtout quand on gère une infrastructure avec plusieurs serveurs.

Quel est le rôle du fichier inventory ?

Le fichier inventory.ini est le fichier qui indique à Ansible quelles machines il doit gérer. Il contient les adresses IP des serveurs regroupées par catégorie comme web, db ou security, et peut aussi contenir des variables communes comme le nom d'utilisateur SSH à utiliser pour se connecter.

Quelle différence entre copy et template ?

Le module copy envoie un fichier tel quel sans aucune modification, le contenu reste identique à la source. Le module template lui traite le fichier comme un modèle Jinja2 avant de l'envoyer, ce qui permet d'y insérer des variables dynamiques comme le nom du serveur ou son adresse IP. On utilise copy pour des fichiers fixes et template quand le contenu doit changer selon la machine cible.

Pourquoi séparer les rôles ?

Séparer les rôles permet d'avoir un projet organisé et facile à maintenir. Chaque rôle gère un seul service de manière indépendante, ce qui permet de le réutiliser dans d'autres projets sans tout réécrire. Si un problème survient sur la configuration web par exemple, on sait exactement où chercher sans toucher au reste du projet.

Conclusion : Ce TD nous a permis de comprendre concrètement comment Ansible simplifie la gestion d'une infrastructure en automatisant des tâches répétitives de manière fiable et cohérente. Nous avons appris à structurer un projet Ansible proprement avec des rôles séparés, un inventaire clair et un playbook principal. L'utilisation des templates Jinja2 a également montré comment rendre les déploiements dynamiques et adaptables à chaque serveur cible.